Richards Britto

Deep Learning & Computer Graphics Engineer

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Education

Master's | Uppsala University

Sept 2021 – Nov 2023 — Uppsala, Sweden

- Major: Image Analysis and Machine Learning

Bachelor's | National Institute of Technology Trichy

Sept 2016 – May 2020 — Tiruchirappalli, India

- Major: Electronics and Communication Engineering
- Minor: Humanities and Social Sciences (Economics)
- GPA: 8.01

Work Experience

AI & 3D Software Developer | Inter IKEA Group

August 2024 – CURRENT — Älmhult, Sweden

- Built AI-driven 3D content pipelines using **Vertex AI** and **Freepik** APIs, with models like **Qwen3-VL** and **Gemma4**, enabling template-based generation of 100k+ assets while reducing rendering costs and manual effort.
- Developed key components of an AI platform on **Azure**, including agent-based systems and a custom RAG pipeline, delivering production-ready solutions.

Machine Learning Engineer | Litigence

May 2024 – July 2024 — Oslo, Norway

- Fine-tuned LLMs with **DPO** for patent analysis, suggesting edits for novelty and detecting existing overlaps.

Synthetic Data Generation | Vidhance AB

Sept 2022 – Jan 2023 — Uppsala, Sweden

- Built **Blender Python API** tool simulating real-world camera effects and sensor motion, saving 20+ hrs/week; collaborated in **agile team** to generate videos with depth map, flow map, and customizable environments.

Academic Research

iNPH Diagnosis using DL (Masters Thesis) |

Feb 2023 – Sept 2023 — Uppsala University, Sweden

- Estimated 3D biomarker voxels with sub-mm accuracy via **nn-UNet**, aligned MRIs to Talairach space, and computed Evan's Index using FastSurfer on 3090-Ti (Linux).

Occlusion Aware Origami Pose Estimation |

May 2020 – April 2021 — NUS, Singapore

- Achieved 97% pose estimation accuracy using a custom **Mask-RCNN+cGAN** pipeline combining RGB-D for depth and 3D keypoint generation.

Technical Skills

- **PL :** Python, C/C++, LaTex, HTML/CSS
- **Library :** PyTorch, OpenCV, Open3D, NumPy, Pandas, Matplotlib, MONAI, scikit-learn, SciPy, PyGame, bpy
- **Dev Tool :** Git, CUDA, SSH, venv, VSCode, Linux, Azure, ARCore, Jiira, Confluence, ComfyUI, Miro
- **Software :** MATLAB, Blender, 3DS Max, CARLA

Project Work

3D Object Detection using LiDAR |

Feb 2024 – Mar 2024 — Uppsala, Sweden

- Implemented SFA3D algorithm for 3D object detection on the **KITTI dataset**, utilizing Bird's Eye View representation of LiDAR point cloud data. Employed **Keypoint-FPN** architecture with **ResNet 18** backbone.

MIP Control and Dynamcs |

Dec 2023 – Feb 2024 — Uppsala, Sweden

- Designed a control system for Mobile Inverted Pendulum in **MATLAB**, incorporating **Extended Kalman Filters** for state estimation and **PD controllers** for dynamic stabilization, resulting in 25% increase in system stability.

NeRF and Gaussian Splatting |

Dec 2023 – Jan 2024 — Uppsala, Sweden

- Employed Blender's Python API to generate a synthetic dataset containing ∼150 RGB images, camera parameters, along with meta data, then trained **NeRF** and **Gaussian Splatting** to model volumetric scenes.

Monocular ORB-SLAM |

Nov 2023 – Dec 2023 — Uppsala, Sweden

- Executed **6 DoF** camera pose estimation and 3D scene reconstruction utilizing the TUM RGBD dataset.
- Leveraged the **PnP** algorithm for pose estimation, refined results through **bundle adjustment**, and applied **Bags-of-Words** for loop closure, 0.12 RMSE was achieved.

Ray Tracing and Rasterization |

Sept 2023 – Oct 2023 — Uppsala, Sweden

- Engineered rasterization, ray tracing techniques in C++ including motion blur, textures, and lights, achieved a 40% rendering speed boost using BVH.

Autonomous Driving in CARLA | |

Jan 2023 – Feb 2023 — Uppsala, Sweden

- Implemented a **CNN** based **DQN** to control agents using high-dimensional sensory inputs like vision and velocity.